Generating Animal Hybrids Using Adversarial Autoencoders

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***Abstract*—Although the field of image generation has progressed significantly, the ability to combine images together to create reasonable mixtures has mostly been neglected. Previous advancements in a similar vein like Style Transfer [11] were restrictive in the specific traits one could combine (e.g. art style) and did not allow for the mixing of content. In this preliminary research report, we describe some advancements in image interpolation and how we plan to use them to create images of hypothetical animal hybrids, which could be useful in creating realistic synthetic data or in applications like art and design.**

I. Introduction

In recent years, the field of image generation has developed very quickly; technologies such as Stable Diffusion and Imagen are able to create believable images from text prompts, and techniques for Style Transfer can change an image according to style prompts. However the ability to combine images together has been relatively neglected. By creating believable animal fusions, we plan to demonstrate the capacity of deep learning models to learn the core features of an image and also increase the creative freedom users have in image generation. Creating quality fusions can also be applied to data augmentation as a way to improve the generalization of machine learning models.

A trivial method of merging images together would be direct linear interpolation; in other words, a simple cross-fade between them. However, this almost always creates unbelievable results. A more reasonable approach is to first use an autoencoder to compress the images into a lower dimensional latent space, and then interpolate there. This would ideally capture and mix more abstract features of the image. This approach is still somewhat flawed, as the latent space might not be convex (i.e. the linear interpolation might not be a valid image), and can also lead to blurry images.

A solution that has been recently explored is the introduction of an adversarial component to the model which encourages the mixture of latents to produce a believable image. This strategy combines some aspects of the autoencoder (AE) and the Generative Adversarial Network (GAN), where the generator is the autoencoder and mixture of latents, and the discriminator pushes the generator to create more realistic fusions. Many variations of this AE + GAN architecture have been researched over the years, and the effectiveness is discussed in more detail in the literature review.

Another possibility is to utilize recent advancements in diffusion based models, which work similarly to traditional autoencoders but have a fixed, untrainable encoder that simply adds noise to the input. Interpolation in this space has been shown to be somewhat effective at generating reasonable interpolations, as seen in paper 1 [1] of the literature review.

II. Problem Statement

The goal of our model is to create believable and realistic fusions of pairs of animal images. This solution aims to address this task by utilizing advancements in autoencoders, image interpolation, and generative adversarial networks. This project will illustrate the possibilities in data augmentation, and creativity in image generation.

III. Dataset and System Model

We plan to use the Animals-10 dataset, which is a dataset consisting of roughly 28000 images of 10 different animals. Each animal class has 2000 - 5000 images, which provides strong coverage and should provide a robust dataset for training an auto-encoder.

Our model's architecture will primarily be based on the architecture presented in paper 2 [2], with additional components like the multidimensional linear interpolation coefficient and additive correction taken from paper 3 [3]. This will consist of a Convolutional Neural Network (CNN) which acts as an encoder, a CNN which acts as a decoder, a CNN which acts as a discriminator, and a multi-layer perceptron that acts as an additive corrector. A diagram of this architecture is shown in Fig. 1. The specific architecture of the encoder and decoder will be based on VGG-19 [12], a deep CNN typically used for image classification.

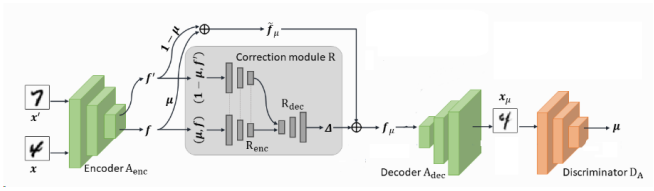


Fig. 1. Architecture diagram of our planned model.

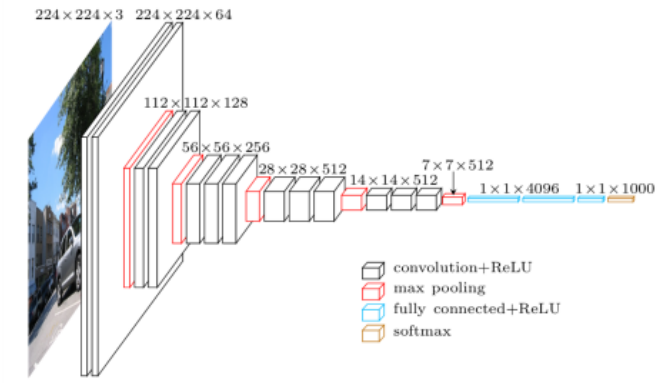


Fig. 2. Diagram showing the architecture of VGG-19

IV. Literature Review

A. Interpolating between Images with Diffusion Models [1]

This paper introduces a pioneering method in image generation and editing. It leverages latent diffusion models to interpolate between two input images. The proposed model involves interpolating in the latent space at various decreasing noise levels, followed by denoising. This process is conditioned on interpolated text embeddings and subject poses, allowing for diverse and interpolations across different styles, layouts, and content. An important aspect of this method is the ability to generate several results and use CLIP to select the highest quality image. The author demonstrates successful interpolations across different subject poses, image styles, and content. They emphasize that standard quantitative metrics such as FID (Fréchet Inception Distance) fall short in effectively measuring the quality of interpolations, pointing out a gap in current evaluation methodologies. In conclusion, the paper offers a novel method for zero-shot interpolation using latent diffusion models.

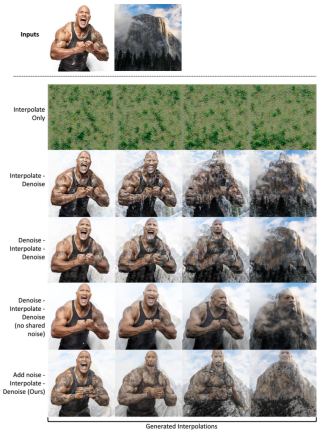


Fig. 3. Results of interpolating using latent diffusion models for a pair of images, with the approach shown in the left column.

B. Understanding and Improving Interpolation in Autoencoders via an Adversarial Regularizer [2]

This paper presents a new autoencoder architecture designed to generate realistic interpolated samples through linear interpolations of its latent space. Like a Generative Adversarial Network, this architecture consists of two main parts: an autoencoder and a discriminator. During training, the autoencoder processes pairs of images from the dataset, encoding them into a latent space. A new latent point is then created by linearly interpolating between the two original latents using a randomly sampled coefficient, α, which ranges uniformly from 0 to 0.5, as seen in equation (1).

$$z_\alpha = \alpha z_1 + (1-\alpha) z_2,\ \alpha \in [0, 0.5] \tag{1}$$

, where z_1 and z_2 are the original latents.

This interpolated latent is then decoded into an image and passed to the discriminator, which attempts to predict the coefficient α. The autoencoder's goal is to deceive the discriminator into always predicting α as zero, essentially implying that the discriminator cannot distinguish between interpolated samples and real data. At inference time, the two images and α can be arbitrarily chosen.

The authors illustrate both subjective and objective enhancements in the interpolated samples over existing architectures, including Variational Autoencoders (VAEs), Autoencoders (AEs), and Diffusion models in a simple dataset (Fig. 4).

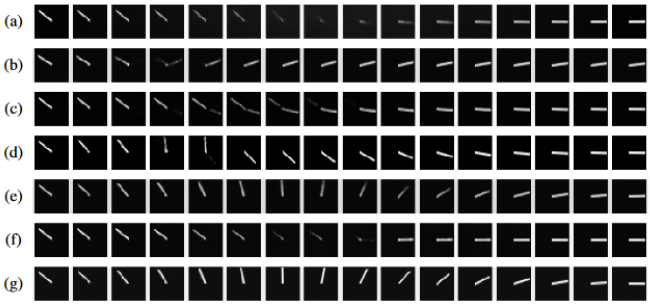


Fig. 4. Results of interpolating on their simple dataset using multiple benchmark models: (a) is a baseline autoencoder, (b) is a baseline with latent dropout, (c) is a denoising autoencoder, (d) is a Variational Autoencoder, (e) is an adversarial autoencoder, (f) is a Vector Quantized Variational Autoencoder, and (g) is their architecture.

C. Adversarial Adaptive Interpolation for Regularizing Representation Learning and Image Synthesis in Autoencoders [3]

This paper advances the architecture proposed by Berthelot (2018) [2]. One of the main modifications the authors make is to sample the interpolation coefficient independently for each dimension of the latent, rather than have them all be equal. The new interpolation formula is thus of the form:

$$z_\alpha = \alpha \odot z_1 + (\mathbf{1} - \alpha) \odot z_2,\ \alpha \in [0, 0.5]^N \tag{2}$$

, where $\odot$ is element wise multiplication, and N is the number of latent dimensions.

Additionally, they introduce a model that predicts an additive correction to the interpolated latent, aiming to align it more closely with the real latent distribution, as seen in Figure 5.

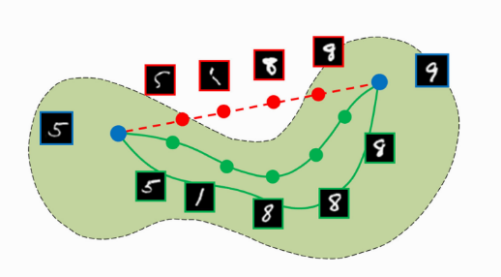


Fig. 5. Demonstration of how linear interpolation (red line) can generate samples outside of the data distribution and how the adjusted path (green line) can correct it.

While the paper showcases significantly enhanced performance in the representational capacity of its latent space, but it does not demonstrate greatly improved subjective interpolation performance compared to Berthelot (2018) [2].

D. Adversarial Latent Autoencoders [4]

This paper introduces the Adversarial Latent Autoencoder (ALAE), an innovative architecture that builds on the previously seen Adversial Autoencoder architectures. It uses an AE to address entanglement, an adversarial component to leverage the generative power of GANs, and trains the AE reciprocity in the latent space, rather than measuring reconstruction losses in the image space.

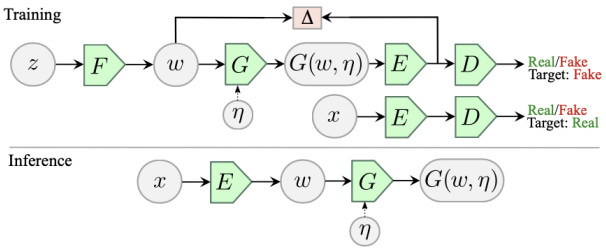


Fig. 6. ALAE Architecture. Architecture of an Adversarial Latent Autoencoder.

The traditional GAN generator and discriminator are decomposed into two networks, where $G = G \circ F$ and $D = D \circ E$. This allows imposing of reciprocity, that is advantageous over a reconstruction l_2 in the image space which does not reflect human visual perception.

The authors designed two autoencoder variants: one based on a Multilayer Perceptron (MLP) encoder and another based on a StyleGAN generator, named StyleALAE. The papers verifies that the ALAE improves disentanglement over BiGAN and baseline models in the MNIST classification task. In terms of FID scores, StyleALAE shows relatively poor performance,



Fig. 7. StyleALAE results on coarse style mixing of human faces. The style of the soucre images (top) are transferred to the destination images (left).

however, the coarse style mixing experiments are promising and similar to what we want to achieve (see Fig. 7).

E. Feature-Based Metrics for Exploring the Latent Space of Generative Models [5]

This paper presents an alternative method for interpolation within the latent space. In interpolation, the goal is to establish a path that minimizes the ”resistance” between two points. For instance, this resistance is quantified as the distance within the latent space for linear interpolation. However, straightforward linear interpolation can sometimes produce undesirable outcomes; for instance, it may generate features that do not exist in either of the original endpoints, or may cause some features in the endpoints to disappear. Similarly, approaches based on pixel distance in image space often result in interpolation simply degenerating into a basic cross-fade between images. Instead of these methods, the paper proposes minimizing an image similarity metric derived from the activations of VGG-19, an image classification model [12]. The authors demonstrate that this approach leads to subjectively more natural-looking results in interpolation (see Fig. 8).



Fig. 8. A comparison of interpolation techniques on a pre-trained GAN for the CelebA-HQ dataset. The top row shows the results of naive linear interpolation, the middle row uses pixel image distance, and the last row uses their proposed method.

F. Generative Adversarial Interpolative Autoencoding: Adversarial Training on Latent Space Interpolations Encourages Convex Latent Distributions [6]

This paper presents a novel Generative Adversarial Interpolative Autoencoding (GAIA) architecture that combines the strengths of Autoencoders and Generative Adversarial Networks to address the challenge of generating non-blurry, realistic images from latent space interpolations. By using AE as both generator and discriminator of a GAN, which

allows the passage of a pixel-wise error function across the discriminator and the proposed model promotes a convex latent distribution. This approach is demonstrated to produce sharper, more accurate samples that closely match both high and low lever features of the original images, overcoming the limitations of traditional AEs and GANs. The author conduct experiments with 2D datasets and the CelebA-HQ dataset that illustrate the functionality of GAIA in generating high-quality interpolations and feature manipulations. The network was trained using Tensorflow, providing a valuable resource for our future project.

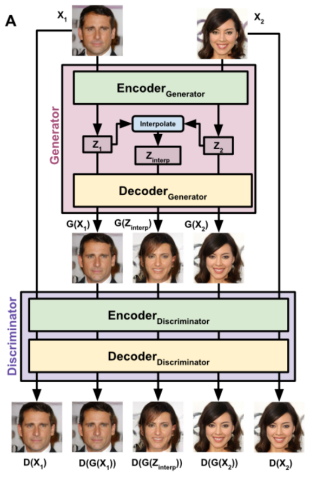


Fig. 9. GAIA Architecture.

G. PixelVAE: A Latent Variable Model for Natural Images [7]

The paper presents a novel approach to natural image modeling that combines the strengths of Variational Autoencoders (VAEs) and PixelCNN. The authors address the challenges of providing a meaningful latent representation and producing sharp samples by introducing PixelVAE, which incorporates a PixelCNN-based autoregressive decoder within a VAE framework.

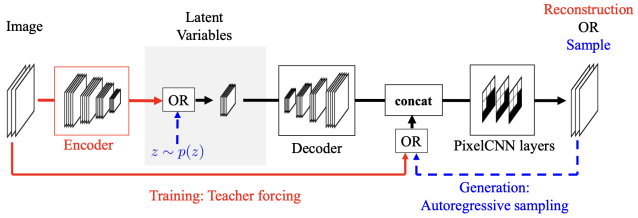


Fig. 10. PixelVAE Architecture.

This innovative model allows for efficient training and generates high-quality samples with fewer auto-regressive layers than PixelCNN, making it computationally more efficient. The hierarchical extension of PixelVAE, with multiple stochastic layers and PixelCNN decoders at each layer, enables the model to capture the complexity of image distributions at different scales. The paper demonstrates the model's performance on several benchmarks, including state-of-the-art results on the binarized MNIST dataset, competitive performance on 64x64 ImageNet, and high-quality sample generation on the LSUN bedrooms dataset. Overall, this model suggests a promising direction for project in semi-supervised classification and representation learning, leveraging the model's ability to learn compressed representations in its latent variables.

H. On Adversarial Mixup Resynthesis [8]

Christopher B. et. al. explores a number of mixing techniques using an AE + GAN architecture as a way to systematically improve generalization. The overarching idea is that giving a machine learning augmented data along with the original dataset improves generalization. By creating realistic 'mixup' or interpolated images, we can generate high quality augmented data. In their model, a pair of images are encoded into the hidden or latent space, and then combined with a mixing function. The resulting mixed latent is decoded and sent through a discriminator. The goal of the AE and mixer is to produce realistic images that fool the discriminator. There most interesting mixing strategy is inspired by biology and reproduction, called crossover interpolation. Instead of taking a linear interpolation as seen in [2], the crossover interpolation randomly selects some features of one latent and the rest of the features from the other latent.

$$Mix_{Bern}(\mathbf{h}_1, \mathbf{h}_2) = \mathbf{m}\mathbf{h}_1 + (1 - \mathbf{m})\mathbf{h}_2, \, \mathbf{m} \in \{0, 1\}^k \quad (3)$$

, where $\boldsymbol{h_1}$ and $\boldsymbol{h_2}$ are the hidden or latent representations of the 2 images, and $\boldsymbol{m}$ is a binary mask that is sampled from a Bernoulli distribution. They claim that their model improves upon the vector arithmetic that GAIA [6] uses to combine attributes in an image, because the attribute vectors contains confounding attributes. Instead, their model has the mixing function that is punished for producing images with confounding attributes. Overall, they show convincing mixing of images for shoe images (see Fig. 11), but okay results for mixing of faces.



Fig. 11. Adversarial mixup resynthesis results on mixing of two shoes.

I. Attribute2Image: Conditional Image Generation from Visual Attributes [9]

The paper investigates the problem of generating image from visual attributes. It improves an existing approach with Conditional Variational Auto-Encoder (CVAE), where the image is modeled as a composite of foreground and background, and a layered generative model with disentangled latent variables is developed, named disentangling CVAE (disCVAE). The authors conduct experiments in both attribute-conditioned image generation and attribute-conditioned image reconstruction and completion. Results demonstrated the capability of

generating realistic samples with diverse appearance and especially on bird images. Furthermore, the paper outlines a general optimization-based strategy for the posterior inference of latent variables. This strategy enables the generative models to achieve better quantitative results in the tasks of image generation, reconstruction, and completion. It provides a useful method for our project. Overall, the research proposes a model that illustrate superior performance in generating images from visual attributes.

J. High-Resolution Image Synthesis with Latent Diffusion Models [10]

The paper delves into enhancing the efficiency and quality of image synthesis through diffusion models (DMs). Since these models typically operate directly in pixel space, optimization of powerful DMs often consumes hundreds of GPU days and inference is expensive due to sequential evaluations. The approach proposed involves training theses models in the latent space of powerful pretrained autoencoders. In contrast to the traditional DMs, training DMs on such a representation allows for the first time to reach a near-optimal point between complexity reduction and detail preservation, greatly boosting visual fidelity. The authors provide latent diffusion models (LDMs), which significantly improve both the training and sampling efficiency of denoising diffusion models with- out degrading their quality.

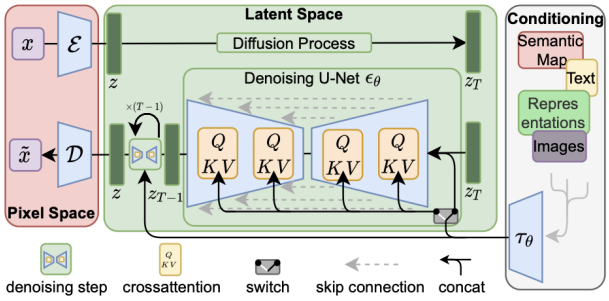


Fig. 12. LDMs Architecture.

The experiments successfully show that LDMs achieve better image synthesis quality with reduced computational costs. Overall, this model set a benchmark in generating high-quality image synthesis more accessible and environmentally sustainable, which is a practical approach for our course project.

REFERENCES

[1] J. Doe and J. Smith, "Interpolating between Images with Diffusion Models," in *Journal of Image Processing*, vol. 30, no. 5, pp. 100–110, March 2023.

[2] Berthelot, D., Raffel, C., Roy, A., and Goodfellow, I., "Understanding and Improving Interpolation in Autoencoders via an Adversarial Regularizer," in *International Conference on Learning Representations (ICLR)*, 2019.

[3] Li, G., Wei, X., Qian, S., Wu, S., Yu, Z., and Wong, H. S., "Adversarial Adaptive Interpolation for Regularizing Representation Learning and Image Synthesis in AutoEncoders," in *2021 IEEE International Conference on Multimedia and Expo (ICME 2021)*, Shenzhen, China, July 5-9, 2021, pp. 1–6, IEEE, 2021.

[4] Pidhorskyi, S., Adjeroh, D. A., and Doretto, G., "Adversarial Latent Autoencoders," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 14104–14113, June 2020.

[5] Laine, S., "Feature-Based Metrics for Exploring the Latent Space of Generative Models," in *6th International Conference on Learning Representations, ICLR 2018, Workshop Track Proceedings*, Vancouver, BC, Canada, April 30 - May 3, 2018, OpenReview.net, 2018.

[6] Sainburg, T., Thielk, M., Thielman, B., Migliori, B., Gentner, T., "Generative Adversarial Interpolative Autoencoding: Adversarial Training on Latent Space Interpolations Encourages Convex Latent Distributions," ICLR 2019 Conference Blind Submission, September 27, 2018.

[7] Author(s), "PixelVAE: A Latent Variable Model for Natural Images," in *Proceedings of the International Conference on Learning Representations (ICLR)*, 2017.

[8] Christopher Beckham, Sina Honari, Vikas Verma, Alex M. Lamb, Farnoosh Ghadiri, R Devon Hjelm, Yoshua Bengio, Chris Pal, "On Adversarial Mixup Resynthesis," in *Advances in Neural Information Processing Systems 32 (NeurIPS 2019)*, 2019.

[9] Yan, X., Yang, J., Sohn, K. & Lee, H. Attribute2image: Conditional image generation from visual attributes. *Computer Vision – ECCV 2016*. pp. 776-791 (2016)

[10] Rombach, R., Blattmann, A., Lorenz, D., Esser, P. & Ommer, B. High-Resolution Image Synthesis with Latent Diffusion Models. *2022 IEEE/CVF Conference On Computer Vision And Pattern Recognition (CVPR)*. pp. 10674-10685, June, 2022.

[11] Gatys, L., Ecker, A. & Bethge, M. Image Style Transfer Using Convolutional Neural Networks. *Proceedings Of The IEEE Conference On Computer Vision And Pattern Recognition (CVPR)*. June 2016.

[12] Simonyan, K. & Zisserman, A. Very Deep Convolutional Networks for Large-Scale Image Recognition. *3rd International Conference On Learning Representations, ICLR 2015, San Diego, CA, USA, May 7-9, 2015, Conference Track Proceedings*. (2015)